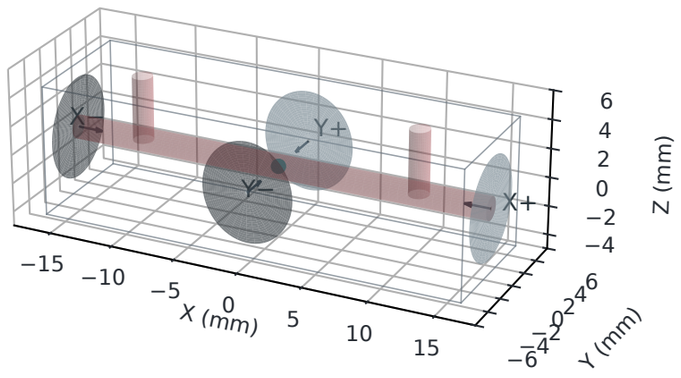
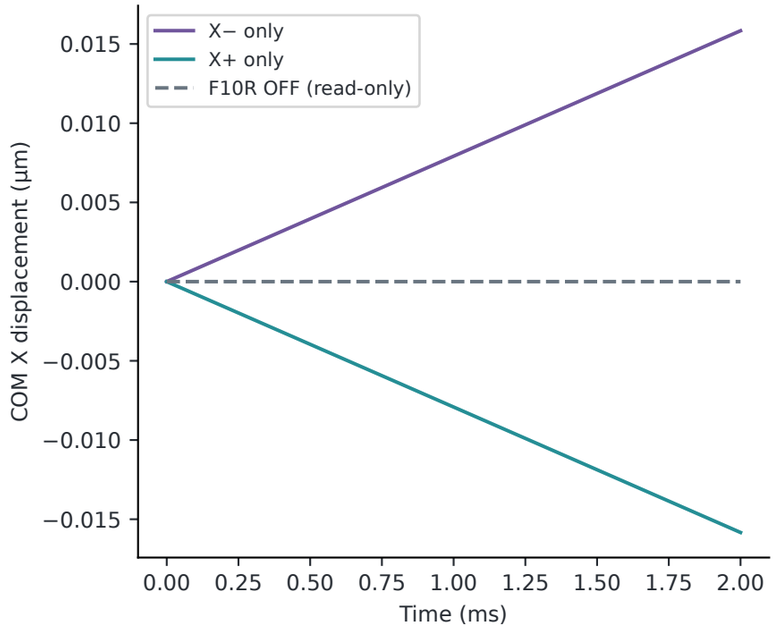
Independent single-PZT direction verification

a Confirmed apparatus geometry

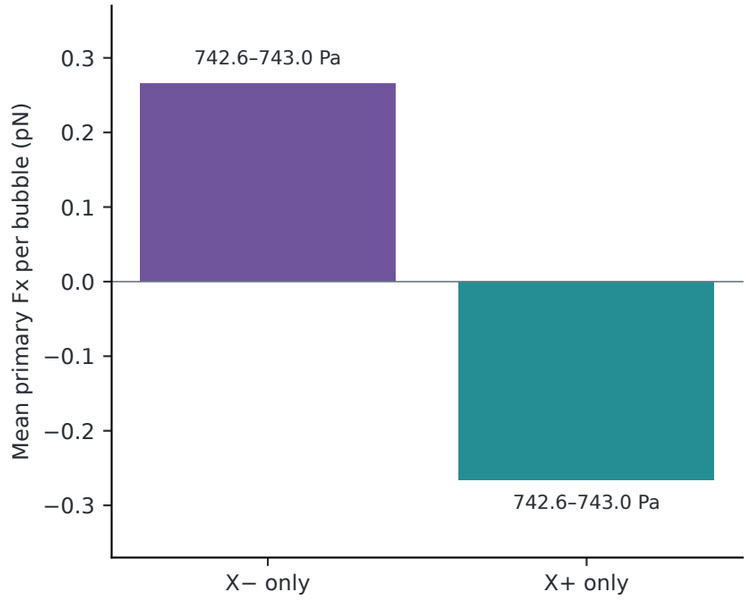


Water-filled cavity; zero flow
Both X ends sealed; 7 mm PZT discs

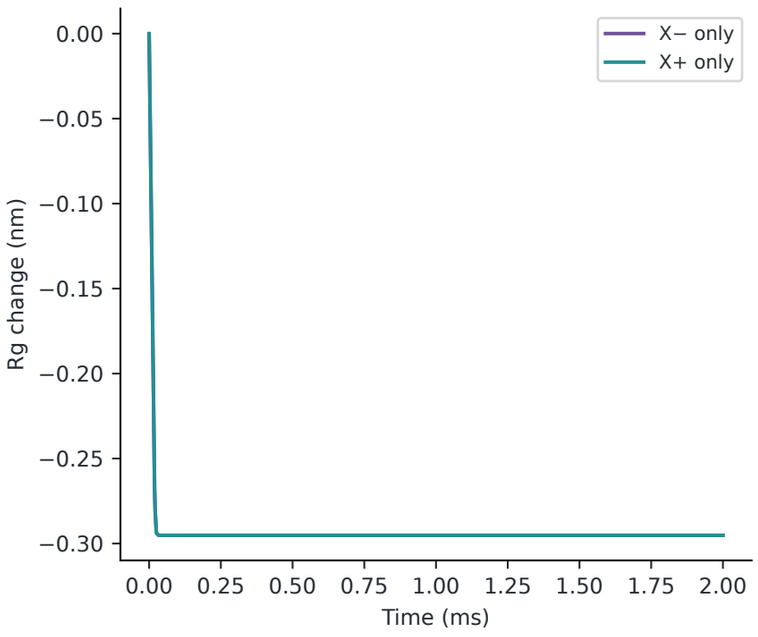
b Independent runs, identical initial state



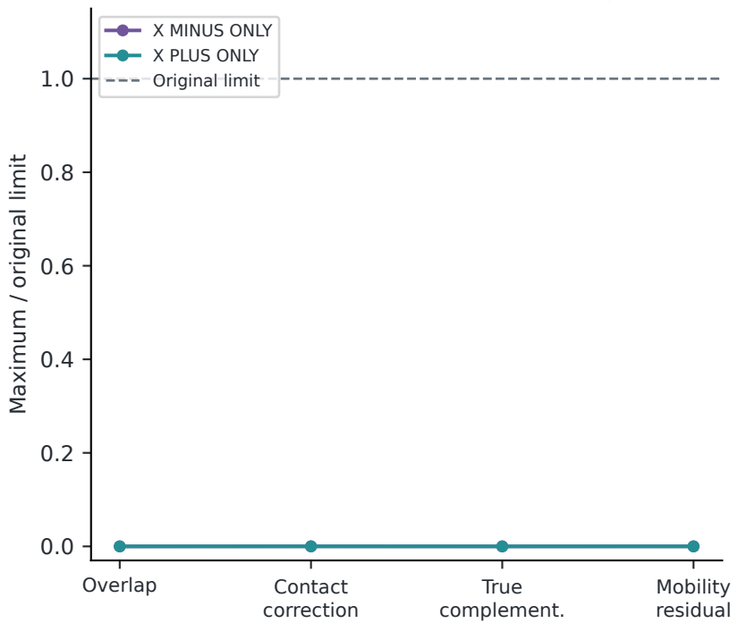
c Measured pressure and force



d Measured shape change



e Contact and numerical quality



f Direction, signal and scope

Each source: 400 Pa-equivalent, 2.5 MHz
All other sources OFF; no background flow
Independent 0.5 ms screens → fresh 2 ms runs

X−: ΔX = +0.01583580 µm
Mean velocity = +7.91790 µm/s
Largest COM numerical difference = 0.000490 nm

X+: ΔX = -0.01583580 µm
Mean velocity = -7.91790 µm/s
Largest COM numerical difference = 0.000490 nm

All five numerical layers: PASS
Connected 32-ID network retained
Small shape change is not physical fusion

Full primary + secondary forces, F2, pair lubrication, hard-sphere SQP and walls · Not extracted from a switched sequence
EXPLORATORY · HOMOGENEOUS_WATER_CONTROL_PROXY · PHYSICS_UNCALIBRATED · training_ready=false